Zero-shot, one-shot and few-shot prompting

1. What You'll Learn in This Section

In this lesson, you'll learn to:

- Classify any prompt as zero-shot, one-shot or few-shot by counting its examples
- Decide when a task needs examples and when it is really a zero-shot task
- Build a few-shot prompt using the task-plus-input-and-output template
- Write a few-shot prompt that sorts customer support tickets into four categories

2. Detailed Explanation

A quick recap of prompting

Prompting is giving instructions or input to an AI model. A prompt is an instruction that explains your requirement to the AI in the most simplified format. The prompt goes to the LLM, and the LLM generates a response.

You may already know the GCCFE framework, one way to structure that input. It is not the only framework, but it is a handy checklist. Its five elements are:

- Goal
- Context
- Constraint
- Format
- Example

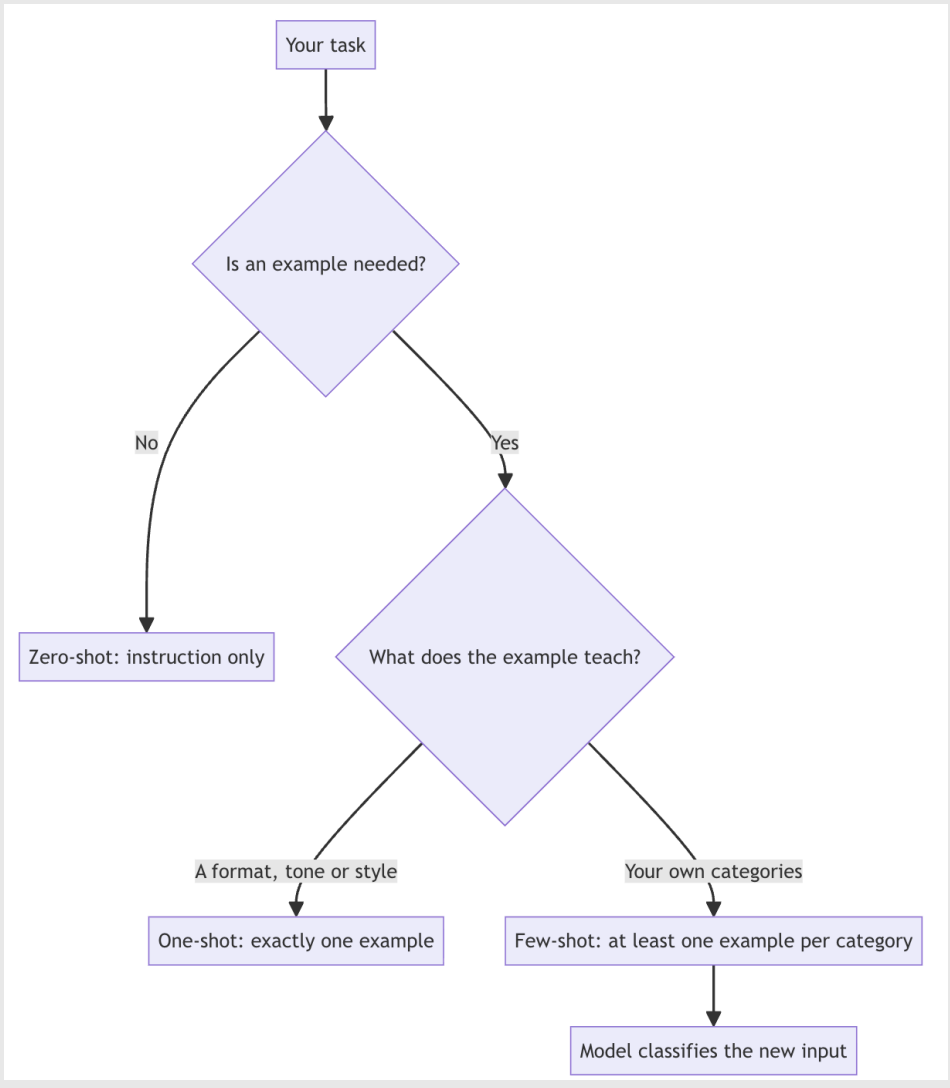
Notice that last element. Whether you include an example depends on the task. Sometimes examples help and sometimes they don't. Either way, the model has to understand the task from your instructions plus any examples you give. This whole lesson is about getting that "Example" choice right.

Three types of prompting by number of examples

At the most basic level, you can sort prompts into three types. The word shot simply means an example.

Type	Number of examples	Meaning
Zero-shot	0	No example is provided because none is needed
One-shot	Exactly 1	One example is provided
Few-shot	A small number, greater than 1	Several examples so the model can understand the pattern

The basic few-shot setup is an instruction plus examples, followed by a new input. The LLM (or any AI model) then generates the output for that new input.



Zero-shot prompting

Zero-shot prompting covers any task where no example or reference is needed. Asking an AI "what is 1 plus 1" is the simplest case. Other everyday zero-shot tasks include:

- Mathematical questions, such as "Please help me with the below mathematical calculation."
- Summarising an article.
- Creating an email summary. Summarisation needs no example.
- Coding requests, such as "Please write a code to count the number of digits in the given number."

That coding case makes the logic clear. You don't supply a sample solution, because if you already knew how to write the code, you wouldn't need to ask.

One-shot prompting

One-shot prompting gives the model exactly one example. Reach for it whenever the output must follow a specific format, tone or style. Here are some typical one-shot requests:

- "Write an email to help me apply for a leave. Use this sample email as an example to understand my tone."
- "I need help with writing a LinkedIn post. This is a sample post which gathered a good amount of interaction from the audience. Please help me create a similar post."
- Writing any post or email in a particular tone.

In each case, one sample is enough to show the model what "my style" looks like.

Few-shot prompting

Few-shot prompting is what you need when the AI must learn your own specifications, and multiple examples are mandatory. A useful rule of thumb: the minimum number of examples usually equals the number of categories, with one example per category. More examples are better.

Use case	Categories	Minimum examples
Spam detection	Spam, not spam	2
Sentiment analysis	Happy, sad, neutral	3

Image classification	Dog, cat, rabbit	3
Temperature readings	High, low, normal	3

For spam detection, the task is "Categorise the given email as spam or not spam." Your prompt can say "Consider this example to understand what is spam" and "Consider this example to understand what is not spam." Then add as many further examples as you can.

Debugging is another good fit. A case such as "Checkout page gives a 500 server error" works well as a few-shot use case.

Here's the big picture. Few-shot is the most useful type of prompt in day-to-day work. Zero-shot and one-shot prompts are direct and need little context. Most real tasks need the extra context that several examples provide.

Why examples help

Examples do more than decorate a prompt. They help in two main ways.

They reduce ambiguity. This is the primary reason. Take the instruction "Please convert this sentence into a professional language." The word "professional" is subjective. What feels professional to one person may not match the level another person expects. Share one example, and the meaning becomes clear.

They convey different elements of the prompt. Each GCCFE element can travel through an example:

- One example can show the output format, such as a reference table when you need tabular output.

- Another example can describe the task in a detailed manner.
- Another example can show the do's and don'ts that every task has.

Examples act as instructions

An example is not just a data point. It can also act as an instruction. When you label "this tweet is positive" and "this tweet is negative", you tell the model which direction counts as positive. With positive, negative and neutral classes, each labelled example instructs the model on how to classify.

Some things simply can't be conveyed without examples. Try defining "positive" in the abstract. It's not really possible until you show a few cases.

The general template for few-shot prompts

Every few-shot prompt follows the same structure:

1. Explain the task.
2. Give example 1: its input and its output.
3. Give example 2: its input and its output.
4. Continue for further examples.

Here is the template applied to tweet sentiment:

```
Task: I need help to classify the tweets as positive, negative or neutral.

Example 1
Input: The movie was fantastic.
Output: Positive

Example 2
Input: The movie was a waste of time.
Output: Negative
```

```
Example 3
Input: I watched a movie.
Output: Neutral
```

The third input is neither positive nor negative in context, so it gets the neutral label. A similar review version pairs "Absolutely fantastic film, highly recommended" with positive and "A complete waste of time and money" with negative.

Watch out for: leaving out the outputs. The output is mandatory for every example, not optional. Without labelled outputs, the model cannot learn what counts as positive or negative.

When a "classification" task is actually zero-shot

This one trips people up. Consider the prompt "Classify the following ML algorithms as supervised or unsupervised." Suppose you add linear regression as supervised and K-means as unsupervised. That looks like few-shot prompting, because examples are given. Logically, though, it is a zero-shot task.

Why? The labels are known facts. Linear regression and logistic regression are supervised, and nobody gets to decide that. There's no subjectivity, so no example is needed. It's like stating that 2 plus 2 equals 4. That is mathematical, and the model answers it correctly without examples.

Keep this rule in mind: examples are needed when there is subjectivity, meaning you decide how something should be categorised.

Worked exercise: customer support ticket classification

Customer support is a classification problem, and most customer support systems are becoming AI-driven. Imagine a company that handles four ticket categories:

1. Product-related queries
2. Account-related queries
3. Delivery-related queries
4. Payment-related queries

Your job is to write a few-shot prompt that sorts tickets into these four categories. The trick is to pick deliberately ambiguous examples. Each one could fit more than one category, so your labels teach the model how to treat those borderline cases.

```
I need help with categorisation of the support tickets into the following four segments: product related, account related, delivery related and payment related.
Please consider the below examples to get a proper context about what type of ticket belongs to which segment.

Example 1
Input: The payment was deducted, but the order got cancelled and does not appear in the order history.
Output: Payment related

Example 2
Input: My order has not arrived yet, even though it is marked as delivered.
Output: Delivery related
```

The prompt continues with two more examples:

```
Example 3
Input: I have received a different product than what was mentioned.
Output: Product related

Example 4
Input: I am unable to edit my email address on the app.
Output: Account related
```

Now look at the choices behind each label:

- Example 1 could be payment or delivery related. You choose payment.
- Example 2 could fit several segments. You choose delivery.
- Example 3 could be delivery or product related. You choose product, because the product specifications differ from what was ordered.
- Example 4 is a standard account issue. It is neither payment nor delivery.

This is subjectivity in action. Your labels decide how the model treats each ambiguous case, so the examples are how you tell it which way you lean.

Want more practice inputs? Try labelling these yourself: "I was billed on my credit card", "Login is not working" (account related), "Money deducted from my card but order not placed", and "Delivered before time."

Quick self-check

Test yourself on the core definitions:

Question	Answer
What is prompting?	Giving instructions or input to an AI model
What is few-shot prompting?	Giving a few examples along with the instructions
What is zero-shot prompting?	Prompting without providing any example

What is one-shot prompting?	Providing exactly one example
What is the purpose of examples in few-shot prompting?	To show the model the expected pattern of behaviour

For that last question, watch for tempting wrong answers like reducing the size of the model, increasing computer storage, or deleting previous knowledge. Examples do none of those things.

3. Key Takeaways

- Prompts fall into three types by example count: zero-shot (none), one-shot (exactly one) and few-shot (a small number, more than one).
- Use one-shot when you need a specific format, tone or style. Use few-shot when the model must learn your own categories, with at least one example per category.
- Examples reduce ambiguity, convey prompt elements like format and do's and don'ts, and act as instructions in their own right.
- A few-shot prompt states the task, then gives input and output pairs. Every example needs its output.
- Examples matter only when categorisation is subjective. Factual tasks, like labelling linear regression as supervised, stay zero-shot.

Mental model: Think of examples as labelled answers you show the model so it learns the pattern. You only need them when the right answer is your call, not a known fact.